LooseEnds

The knot tying vibe and code

AI-GENERATED CODE Upload Clear 461 20,0

```
1   def get_user_discount(us
2       discount = 0
3
4       # Bug 1: Security/Lo
5       if user_role == "adm
6           discount = 0.5
7       elif user_role == "m
8           discount = 0.2
9
10      # Bug 2: Logic flaw
11      if cart_total > 100:
12          discount += 0.1
13
14      return cart_total *
```

LANGUAGE Python

Explain simply

PROGRAM CONTEXT

E-commerce checkout engine. Handles discount calculations for users.

CONVENTION RULES

Paste your team's coding standards or .eslintrc rules here (optional). For example: 'All variables must be

Add comments

Respect comments

THE ENGINE

Static & Security Review

Paste your code and the AI will aggressively scan for SQL injection, logic holes, unhandled exceptions, and convention violations. It provides a concrete fix and a unit test for every issue it finds.

Autonomous Fuzzing Loop

Teach the AI (Memory Vault)

Context Features: Toggle "Add comments" to generate docstrings, "Respect comments" to keep your TODOs, and "Auto-indent" to fix messy formatting.

The API key never reaches the browser.

Auto-indent

Run Definitive Engine

ENGINE EXECUTION TIMELINE

Fuzz Pass 1

All 1 functions passed fuzzing.

Final Review

Static analysis complete.

Multiple logic issues: case-sensitive role comparison and missing validation for empty or negative cart totals.

2 medium Score: 80

Press Cmd+K to clear filters

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MEDIUM logic line 5

The role comparison uses case-sensitive string equality, so roles like "ADMIN" or "Member" are not recognized, leading to incorrect discount calculations.

Show Diff Auto-Applied Teach AI

SUGGESTED FIX Copy

```
if user_role.lower() ==
"admin":
    discount = 0.5
```

```
elif user_role.lower() ==
"member" and
is_first_purchase:
    discount = 0.2
```

GENERATED UNIT TEST

```
import pytest
from user_code import
get_user_discount

def
test_case_sensitivity():
    # Expect discount 0.5
for admin but current
code gives 0
    assert
get_user_discount("ADMIN"
, False, 100) == 100 * (1
- 0.5)
    # Expect discount 0.2
for member first purchase
but current code may not
trigger
    assert
get_user_discount("MEMBER
", True, 50) == 50 * (1 -
0.2)
```

MEDIUM missing flow line 11

The function does not validate that cart_total is a non-negative numeric value, allowing negative or non-numeric inputs to propagate and cause unexpected behavior or TypeError.

Show Diff | **Auto-Applied** | Teach AI

SUGGESTED FIX Copy

```
if cart_total > 100 and
cart_total >= 0 and
isinstance(cart_total,
```

```
(int, float)):
    discount += 0.1
```

GENERATED UNIT TEST

```
import pytest
from user_code import
get_user_discount

def
test_cart_total_validatio
n():
    # Negative total
should not trigger extra
discount and should not
crash
    assert
get_user_discount("member
", False, -10) == -10
    # Non-numeric total
should raise TypeError
when compared to 100
    with
pytest.raises(TypeError):

get_user_discount("member
", False, "50")
```